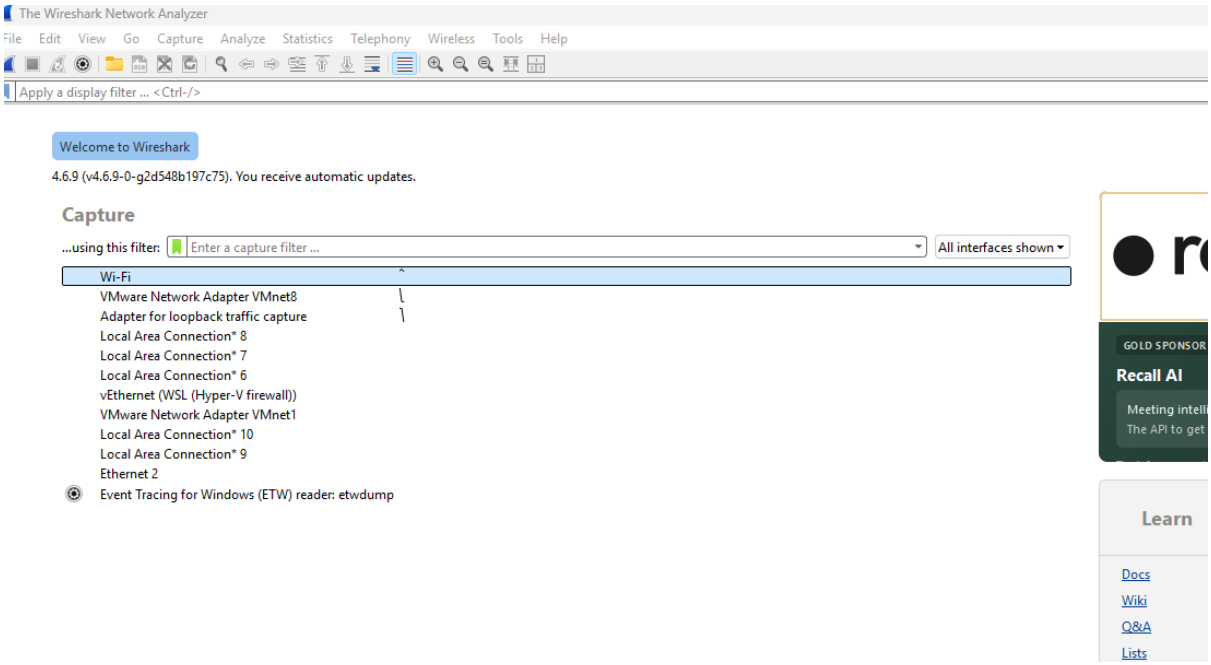
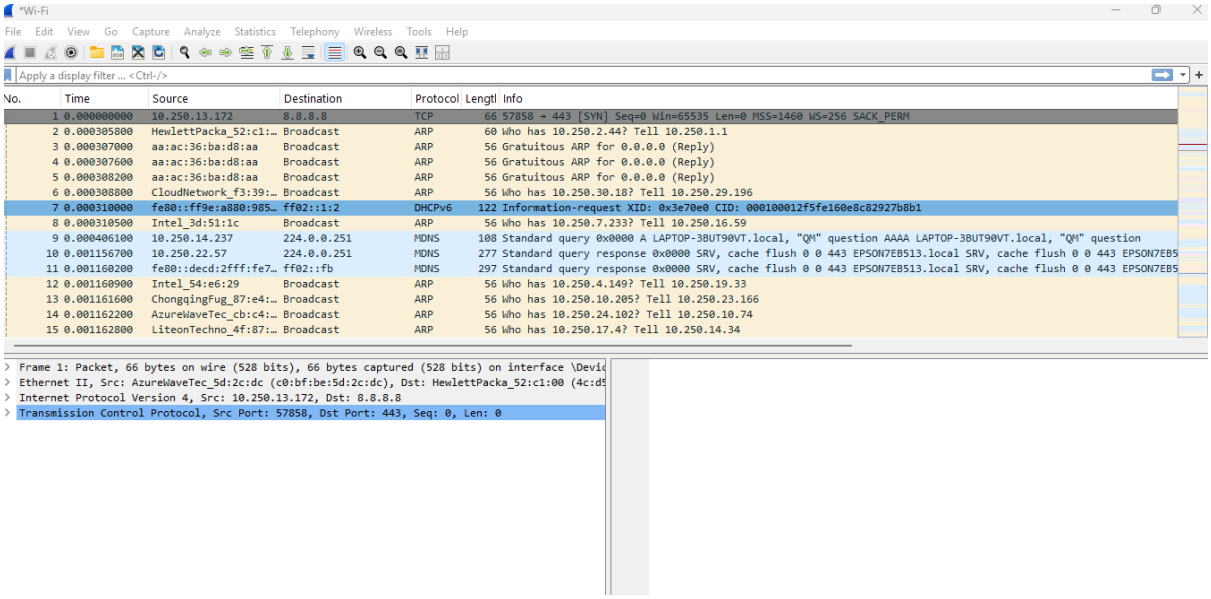
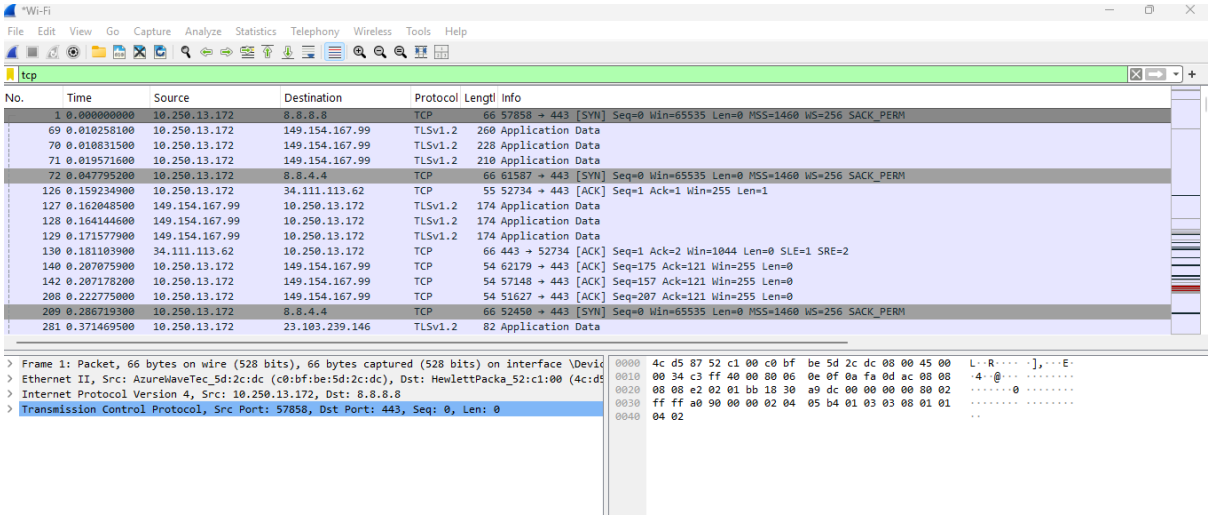
Step1: Open Wireshark and select wifi

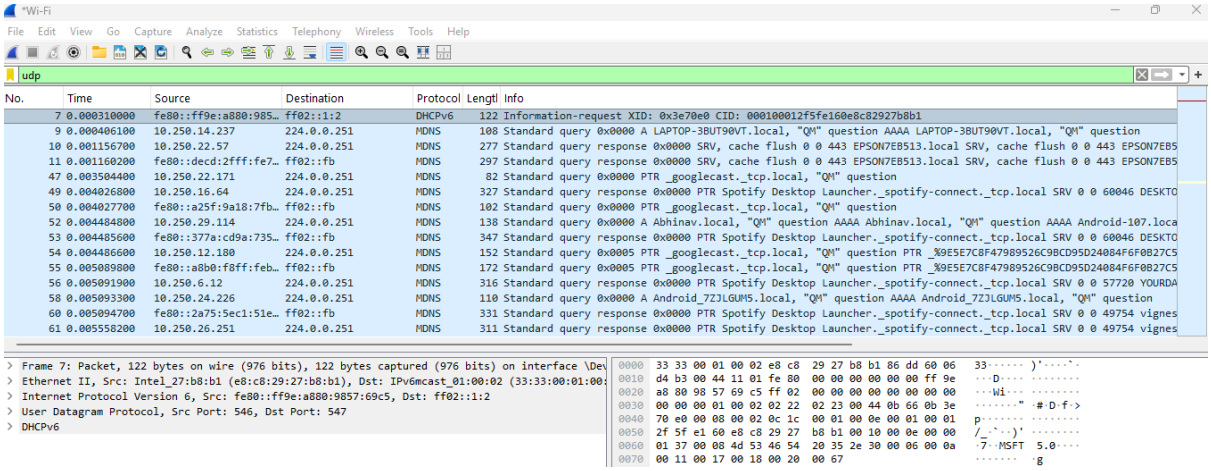


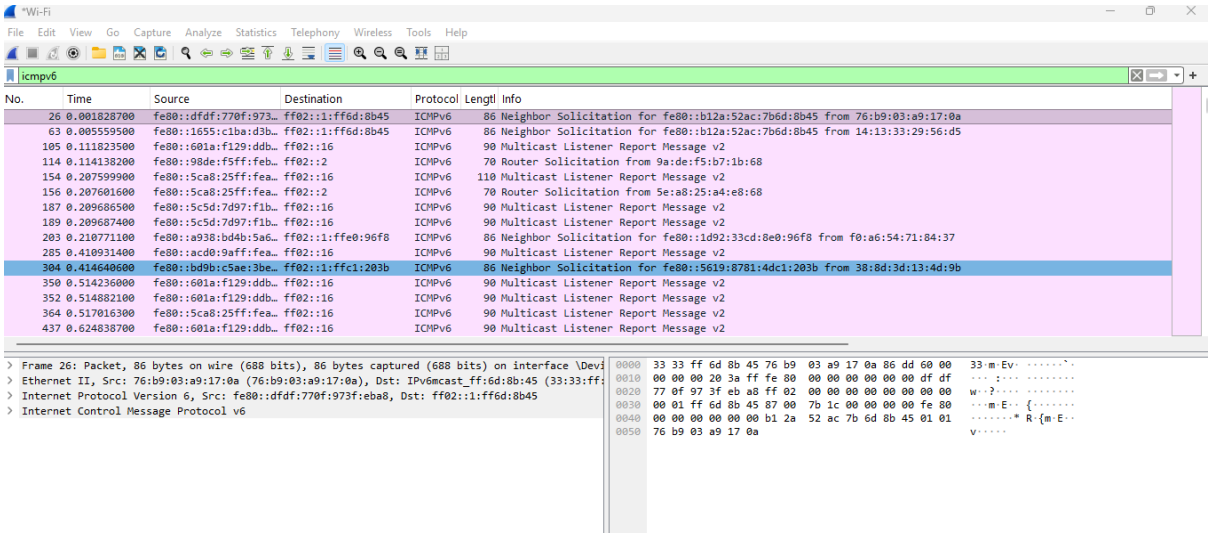
Step 2: Generate network trafficking

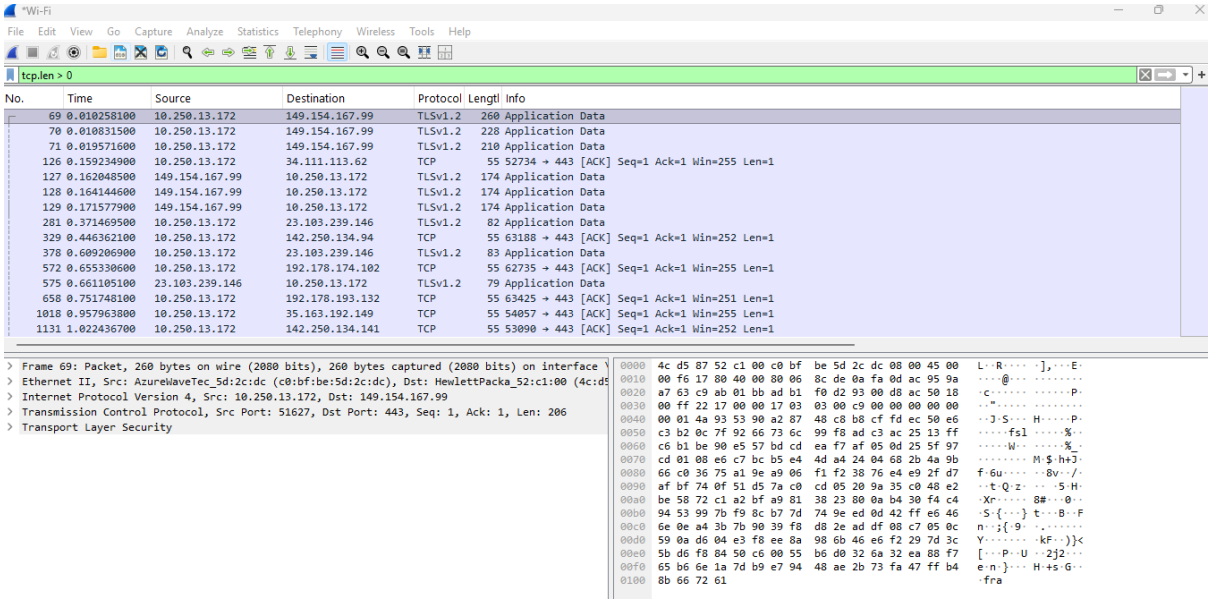


Step3: Apply Display filters


*Wi-Fi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
tcp
No. Time Source Destination Protocol Lengtl Info
1 0.000000000 10.250.13.172 8.8.8.8 TCP 66 57858 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
69 0.010258100 10.250.13.172 149.154.167.99 TLSv1.2 260 Application Data
70 0.010831500 10.250.13.172 149.154.167.99 TLSv1.2 228 Application Data
71 0.019571600 10.250.13.172 149.154.167.99 TLSv1.2 210 Application Data
72 0.047795200 10.250.13.172 8.8.4.4 TCP 66 61587 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
126 0.159234900 10.250.13.172 34.111.113.62 TCP 55 52734 → 443 [ACK] Seq=1 Ack=1 Win=255 Len=1
127 0.162048500 149.154.167.99 10.250.13.172 TLSv1.2 174 Application Data
128 0.164144600 149.154.167.99 10.250.13.172 TLSv1.2 174 Application Data
129 0.171577900 149.154.167.99 10.250.13.172 TLSv1.2 174 Application Data
130 0.181103900 34.111.113.62 10.250.13.172 TCP 66 443 → 52734 [ACK] Seq=1 Ack=2 Win=1044 Len=0 SLE=1 SRE=2
140 0.207075900 10.250.13.172 149.154.167.99 TCP 54 62179 → 443 [ACK] Seq=175 Ack=121 Win=255 Len=0
142 0.207178200 10.250.13.172 149.154.167.99 TCP 54 57148 → 443 [ACK] Seq=157 Ack=121 Win=255 Len=0
208 0.222775000 10.250.13.172 149.154.167.99 TCP 54 51627 → 443 [ACK] Seq=207 Ack=121 Win=255 Len=0
209 0.286719300 10.250.13.172 8.8.4.4 TCP 66 52450 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
281 0.371469500 10.250.13.172 23.103.239.146 TLSv1.2 82 Application Data
Frame 1: Packet, 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface \Devic
Ethernet II, Src: AzureWaveTec_5d:2c:dc (c0:bf:be:5d:2c:dc), Dst: HewlettPacka_52:c1:00 (4c:d5
Internet Protocol Version 4, Src: 10.250.13.172, Dst: 8.8.8.8
Transmission Control Protocol, Src Port: 57858, Dst Port: 443, Seq: 0, Len: 0
0000 4c d5 87 52 c1 00 c0 bf be 5d 2c dc 08 00 45 00
0010 00 34 c3 ff 40 00 80 06 0e 0f 0a fa 0d ac 08 08
0020 08 08 e2 02 01 bb 18 30 a9 dc 00 00 00 00 80 02
0030 ff ff a0 90 00 00 02 04 05 b4 01 03 03 08 01 01
0040 04 02


*Wi-Fi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
udp
No. Time Source Destination Protocol Lengtl Info
7 0.000310000 fe80::ff9e:a880:985… ff02::1:2 DHCPv6 122 Information-request XID: 0x3e70e0 CID: 000100012f5fe160e8c82927b8b1
9 0.000406100 10.250.14.237 224.0.0.251 MDNS 108 Standard query 0x0000 A LAPTOP-3BUT90VT.local, "QM" question AAAA LAPTOP-3BUT90VT.local, "QM" question
10 0.001156700 10.250.22.57 224.0.0.251 MDNS 277 Standard query response 0x0000 SRV, cache flush 0 0 443 EPSON7EB513.local SRV, cache flush 0 0 443 EPSON7EB5
11 0.001160200 fe80::decd:2fff:fe7… ff02::fb MDNS 297 Standard query response 0x0000 SRV, cache flush 0 0 443 EPSON7EB513.local SRV, cache flush 0 0 443 EPSON7EB5
47 0.003504400 10.250.22.171 224.0.0.251 MDNS 82 Standard query 0x0000 PTR _googlecast._tcp.local, "QM" question
49 0.004026800 10.250.16.64 224.0.0.251 MDNS 327 Standard query response 0x0000 PTR Spotify Desktop Launcher._spotify-connect._tcp.local SRV 0 0 60046 DESKTO
50 0.004027700 fe80::a25f:9a18:7fb… ff02::fb MDNS 102 Standard query 0x0000 PTR _googlecast._tcp.local, "QM" question
52 0.004484800 10.250.29.114 224.0.0.251 MDNS 138 Standard query 0x0000 A Abhinav.local, "QM" question AAAA Abhinav.local, "QM" question AAAA Android-107.loca
53 0.004485600 fe80::377a:cd9a:735… ff02::fb MDNS 347 Standard query response 0x0000 PTR Spotify Desktop Launcher._spotify-connect._tcp.local SRV 0 0 60046 DESKTO
54 0.004486600 10.250.12.180 224.0.0.251 MDNS 152 Standard query 0x0005 PTR _googlecast._tcp.local, "QM" question PTR _%9E5E7C8F47989526C9BCD95D24084F6F0B27C5
55 0.005089800 fe80::a8b0:f8ff:feb… ff02::fb MDNS 172 Standard query 0x0005 PTR _googlecast._tcp.local, "QM" question PTR _%9E5E7C8F47989526C9BCD95D24084F6F0B27C5
56 0.005091900 10.250.6.12 224.0.0.251 MDNS 316 Standard query response 0x0000 PTR Spotify Desktop Launcher._spotify-connect._tcp.local SRV 0 0 57720 YOURDA
58 0.005093300 10.250.24.226 224.0.0.251 MDNS 110 Standard query 0x0000 A Android_7ZJLGUM5.local, "QM" question AAAA Android_7ZJLGUM5.local, "QM" question
60 0.005094700 fe80::2a75:5ec1:51e… ff02::fb MDNS 331 Standard query response 0x0000 PTR Spotify Desktop Launcher._spotify-connect._tcp.local SRV 0 0 49754 vignes
61 0.005558200 10.250.26.251 224.0.0.251 MDNS 311 Standard query response 0x0000 PTR Spotify Desktop Launcher._spotify-connect._tcp.local SRV 0 0 49754 vignes
Frame 7: Packet, 122 bytes on wire (976 bits), 122 bytes captured (976 bits) on interface \Dev
Ethernet II, Src: Intel_27:b8:b1 (e8:c8:29:27:b8:b1), Dst: IPv6mcast_01:00:02 (33:33:00:01:00:
Internet Protocol Version 6, Src: fe80::ff9e:a880:9857:69c5, Dst: ff02::1:2
User Datagram Protocol, Src Port: 546, Dst Port: 547
DHCPv6
0000 33 33 00 01 00 02 e8 c8 29 27 b8 b1 86 dd 60 06
0010 d4 b3 00 44 11 01 fe 80 00 00 00 00 00 00 ff 0e
0020 a8 80 98 57 69 c5 ff 02 00 00 00 00 00 00 00 00
0030 00 00 00 01 00 02 02 22 02 23 00 44 0b 66 0b 3e
0040 70 e0 00 08 00 02 0c 1c 00 01 00 0e 00 01 00 01
0050 2f 5f e1 60 e8 c8 29 27 b8 b1 00 10 00 0e 00 00
0060 01 37 00 08 4d 53 46 54 20 35 2e 30 00 06 00 0a
0070 00 11 00 17 00 18 00 20 00 67


*Wi-Fi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
icmpv6
No. Time Source Destination Protocol Lengtl Info
26 0.001828700 fe80::dfdf:770f:973… ff02::1:ff6d:8b45 ICMPv6 86 Neighbor Solicitation for fe80::b12a:52ac:7b6d:8b45 from 76:b9:03:a9:17:0a
63 0.005559500 fe80::1655:c1ba:d3b… ff02::1:ff6d:8b45 ICMPv6 86 Neighbor Solicitation for fe80::b12a:52ac:7b6d:8b45 from 14:13:33:29:56:d5
105 0.111823500 fe80::601a:f129:ddb… ff02::16 ICMPv6 90 Multicast Listener Report Message v2
114 0.114138200 fe80::98de:f5ff:feb… ff02::2 ICMPv6 70 Router Solicitation from 9a:de:f5:b7:1b:68
154 0.207599900 fe80::5ca8:25ff:fea… ff02::16 ICMPv6 110 Multicast Listener Report Message v2
156 0.207601600 fe80::5ca8:25ff:fea… ff02::2 ICMPv6 70 Router Solicitation from 5e:a8:25:a4:e8:68
187 0.209686500 fe80::5c5d:7d97:f1b… ff02::16 ICMPv6 90 Multicast Listener Report Message v2
189 0.209687400 fe80::5c5d:7d97:f1b… ff02::16 ICMPv6 90 Multicast Listener Report Message v2
203 0.210771100 fe80::a938:bd4b:5a6… ff02::1:ffe0:96f8 ICMPv6 86 Neighbor Solicitation for fe80::1d92:33cd:8e0:96f8 from f0:a6:54:71:84:37
285 0.410931400 fe80::acd0:9aff:fea… ff02::16 ICMPv6 90 Multicast Listener Report Message v2
304 0.414640600 fe80::bd9b:c5ae:3be… ff02::1:ffc1:203b ICMPv6 86 Neighbor Solicitation for fe80::5619:8781:4dc1:203b from 38:8d:3d:13:4d:9b
350 0.514236000 fe80::601a:f129:ddb… ff02::16 ICMPv6 90 Multicast Listener Report Message v2
352 0.514882100 fe80::601a:f129:ddb… ff02::16 ICMPv6 90 Multicast Listener Report Message v2
364 0.517016300 fe80::5ca8:25ff:fea… ff02::16 ICMPv6 90 Multicast Listener Report Message v2
437 0.624838700 fe80::601a:f129:ddb… ff02::16 ICMPv6 90 Multicast Listener Report Message v2
Frame 26: Packet, 86 bytes on wire (688 bits), 86 bytes captured (688 bits) on interface \Devi
Ethernet II, Src: 76:b9:03:a9:17:0a (76:b9:03:a9:17:0a), Dst: IPv6mcast_ff:6d:8b:45 (33:33:ff:
Internet Protocol Version 6, Src: fe80::dfdf:770f:973f:eba8, Dst: ff02::1:ff6d:8b45
Internet Control Message Protocol v6
0000 33 33 ff 6d 8b 45 76 b9 03 a9 17 0a 86 dd 60 00
0010 00 00 00 20 3a ff fe 80 00 00 00 00 00 00 df df
0020 77 0f 97 3f eb a8 ff 02 00 00 00 00 00 00 00 00
0030 00 01 ff 6d 8b 45 87 00 7b 1c 00 00 00 00 fe 80
0040 00 00 00 00 00 00 b1 2a 52 ac 7b 6d 8b 45 01 01
0050 76 b9 03 a9 17 0a



Step4: TCP and UDP values

Parameter	**TCP Packet**	**UDP Packet**
Frame Length	533 bytes	100 bytes
Ethernet Header Length	14 bytes	14 bytes
IP Header Length	20 bytes	20 bytes
Transport Header Length	32 bytes	8 bytes
IP Total Length	519 bytes	86 bytes
Payload Length	467 bytes	58 bytes

Step5: Calculate header overhead

Parameter	**Value**
Ethernet Header	14 bytes
IPv4 Header	20 bytes
TCP Header	20 bytes
Application Data	1200 bytes
FCS	4 bytes

Total size including FCS:14+20+20+1200+4=1258

Total overhead:14+20+20+4=58

Payload efficiency:1200/1258*100=95.39%

Step6: Observation table

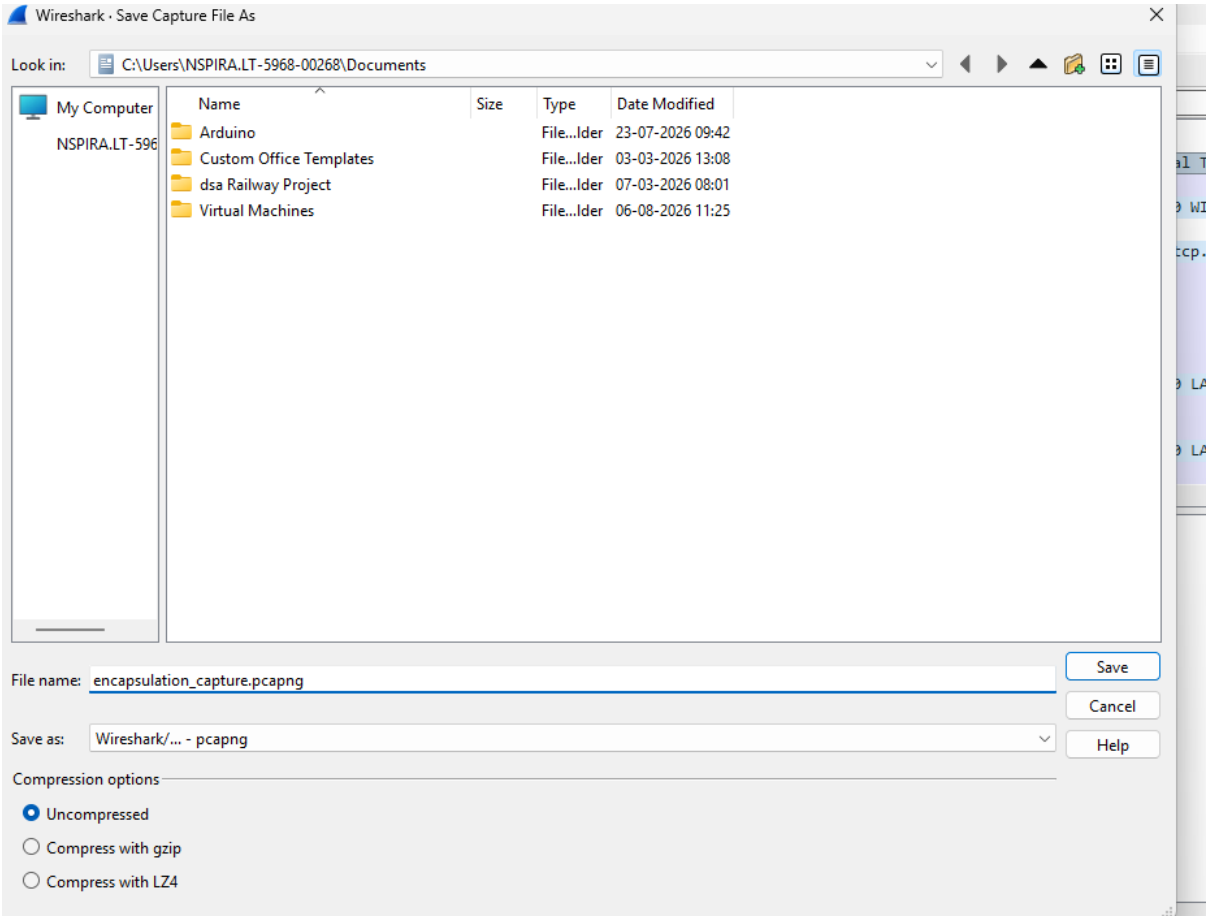
Parameter	TCP Packet	UDP Packet
Frame Length	533 bytes	100 bytes
Ethernet Header Length	14 bytes	14 bytes
IP Header Length	20 bytes	20 bytes
Transport Header Length	32 bytes	8 bytes
IP Total Length	519 bytes	86 bytes
Payload Length	467 bytes	58 bytes

Step7: Save capture file

1. Open Wireshark.
2. Click File → Save As.
3. Choose Desktop.
4. Enter this filename:

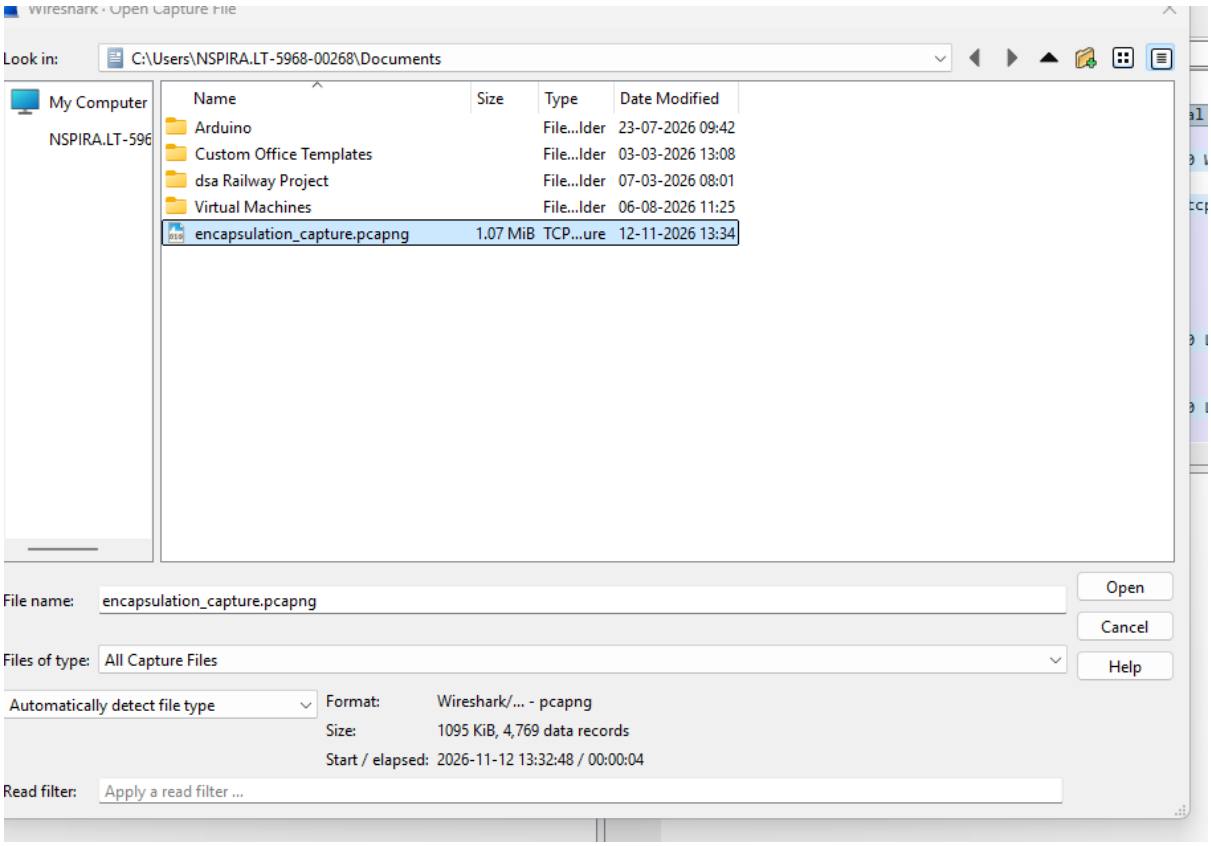
encapsulation_capture.pcapng

5. Keep the format as pcapng.
6. Click Save.



Step 8 – Open the Saved Capture

- Click File →Open
- 2. Select encapsulation_capture.pcapng.
- 3. Click Open.
- 4. Type icmp in the display filter bar.
- 5. Press Enter.



Step9: Answer the questions

1. What is encapsulation?
   Encapsulation is the process of adding protocol headers to data at each network layer before transmission.

2. What is the purpose of the Ethernet header?

It contains source and destination MAC addresses and identifies the type of data being carried.

3. What information is stored in the IP header?

It contains source and destination IP addresses, packet length, TTL and protocol information

4. What is the difference between TCP and UDP headers?

TCP has a larger header with sequence numbers, acknowledgment numbers and control flags. UDP has a simple 8-byte header.

5. Why does each protocol add a header?

Headers provide information needed for addressing, routing and proper data delivery.

6. What is packet overhead?

Packet overhead is the additional information added to application data by network protocols.

7. How does header overhead affect network efficiency?

Higher header overhead reduces the percentage of a packet available for actual application data.